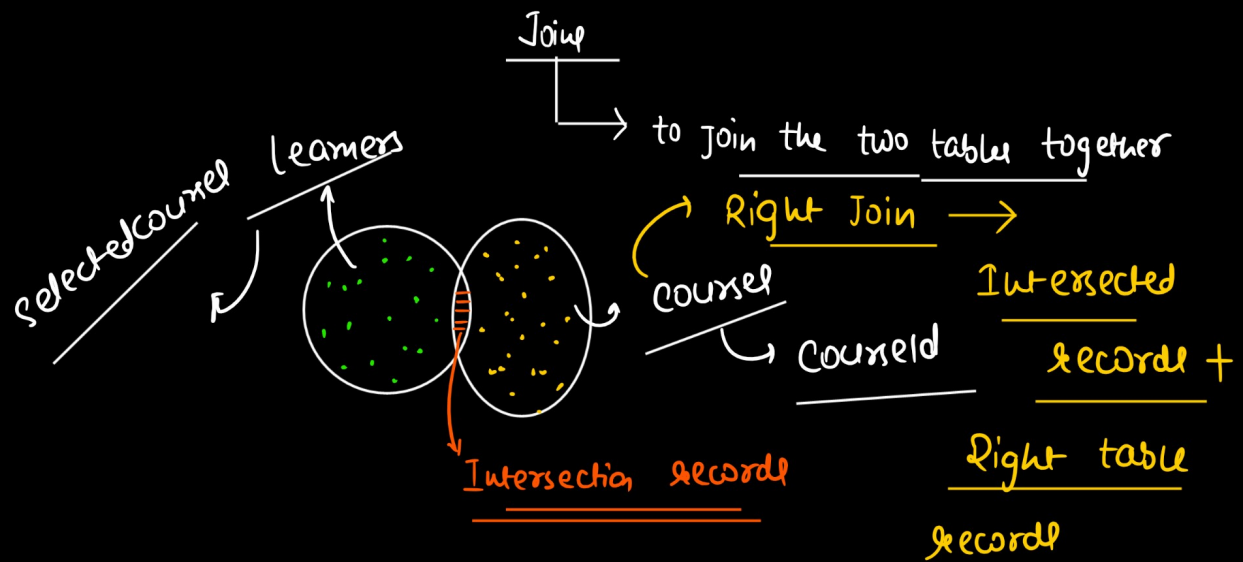




courseID	courseName	courseDurationMonths	courseFee
1	Excel Course	3	1499.00
2	DSA Course	2	4999.00
3	SQL Bootcamp	1	2999.00
4	Data Science	10	14999.00
NULL	NULL	NULL	NULL

LearnerFirstName,

LearnerLastName, selectedCourse, courseName

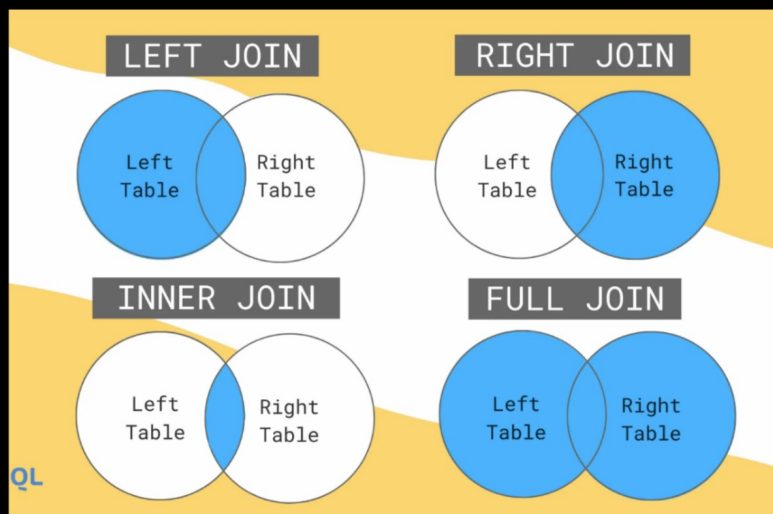
Learner_Id	LearnerFirstName	LearnerLastName	LearnerPhoneNo	LearnerEmailID	LearnerEnrollmentDate	SelectedCourses	YearsOfExperience	LearnerCompany	SourceOfJoining	Batch_Start_Date	Location
1	Akash	Mishra	9998887776	akash@gmail.com	2024-01-21 00:00:00	1	4	Amazon	Linkedin	2024-02-29 00:00:00	Bengaluru
2	Rishikesh	Joshi	9950192388	carjop@gmail.com	2024-03-19 00:00:00	3	2	HCL	Youtube	2024-03-25 00:00:00	Chennai
3	Jeevan	Hegde	9657856732	jeevanhegde@yahoo.co.in	2024-01-15 00:00:00	2	0		Linkedin	2024-01-16 00:00:00	Noida
4	Akhil	George	7689558930	akhil.george.8743@gmail.com	2024-03-13 00:00:00	3	4	Accenture	Community	2024-03-25 00:00:00	Bengaluru
5	Sidhish	Kumar	6475765443	sidhishkumar@gmail.com	2024-01-10 00:00:00	1	4	Meta	Youtube	2024-03-29 00:00:00	Bengaluru
6	NagaSai	Sreedhar	9182937061	saisreedhar2001@gmail.com	2024-03-17 00:00:00	3	4	TCS	Community	2024-03-25 00:00:00	Mumbai

LearnerFirstName	LearnerLastName	SelectedCourses	CourseName
Akash	Mishra	1	Excel Course
Rishikesh	Joshi	3	SQL Bootcamp
Jeevan	Hegde	2	DSA Course
Akhil	George	3	SQL Bootcamp
Sidhish	Kumar	1	Excel Course
NagaSai	Sreedhar	3	SQL Bootcamp

SELECT LearnerFirstName, LearnerLastName, SelectedCourses, CourseName
FROM learners l (left)
INNER JOIN courses c (right)
ON l.SelectedCourses = c.courseID;

LearnerFirstName	LearnerLastName	SelectedCourses	CourseName
Sidhish	Kumar	1	Excel Course
Akash	Mishra	1	Excel Course
Jeevan	Hegde	2	DSA Course
NagaSai	Sreedhar	3	SQL Bootcamp
Akhil	George	3	SQL Bootcamp
Rishikesh	Joshi	3	SQL Bootcamp
NULL	NULL	NULL	Data Science

SELECT LearnerFirstName, LearnerLastName,
SelectedCourses, CourseName
FROM learners l
RIGHT JOIN courses c
ON l.SelectedCourses = c.courseID;



Employee

Self Join

Manager

id	name	salary	managerId
1	Joe	70000	3
2	Henry	80000	4
3	Sam	60000	Null
4	Max	90000	Null

Joe → 70000 > 60000

Henry → 80000 < 90000

Result → Joe

SELECT e.name AS Employee
FROM Employee e
JOIN Employee m
ON e.managerId = m.id
WHERE e.salary > m.salary;

Joe

70000 > 60000

e

id	name	salary	managerId
1	Joe	70000	3
2	Henry	80000	4
3	Sam	60000	Null
4	Max	90000	Null

Self Join

m

id	name	salary	managerId
1	Joe	70000	3
2	Henry	80000	4
3	Sam	60000	Null
4	Max	90000	Null

$e \rightarrow e1$

id	name	salary	managerId
1	Joe	70000	3
2	Henry	80000	4
3	Sam	60000	Null
4	Max	90000	Null

$e.managerId = m.id$

SELECT e.name AS Employee
 FROM Employee e
 JOIN Employee m
 ON e.managerId = m.id
 WHERE e.salary > m.salary;

$70000 > 60000$

$80000 > 90000$

false

$m \rightarrow e2$

id	name	salary	managerId
1	Joe	70000	3
2	Henry	80000	4
3	Sam	60000	Null
4	Max	90000	Null

Inner Join

Right Join

Left Join

self Join

+ Interview

Problem

Subqueries outer query

↳ (Internal query)

17:29:37 UPDATE customers SET AnnualIncome = REPLACE(REPLACE(AnnualIncome, '\$', ''), ',', '') Error Code: 1175. You are using safe update mode and you tried to update a table without a WHERE that uses a KEY column. To disable safe mode, toggle the option in Preferences -> SQL Editor and reconnect. 0.000 sec

always use primary key filter while

doing an update

using WHERE clause

17:34:47 ALTER TABLE customers MODIFY COLUMN AnnualIncome DECIMAL(10,2) Error Code: 1366. Incorrect DECIMAL value: '0' for column '' at row -1 0.015 sec

SELECT CustomerKey, FirstName, LastName, AnnualIncome
FROM customers
WHERE AnnualIncome > (SELECT AVG(AnnualIncome) AS avg_income FROM customers);

internal SQL query

subquery

subquery > FROM → WHERE →

SELECT

Result Grid	Filter Rows:	Export:	W
firstName	lastName	Location	max_salary
Priya	Bhatia	Bengaluru	200000
Suchitra	Behera	Bengaluru	50000
Palak	Bhatt	Noida	90000
Amit	Agarwal	Noida	150000
Avneet	Kaur	Bengaluru	60000
Rashmi	Tanwar	Gurugram	95000
Rohan	Kapoor	Pune	30000
Priya	Sangwan	Pune	100000
Priyanshi	Bhatia	Noida	10000
Atul	Bhatt	Bengaluru	200000

Result Grid						
	empID	firstName	lastName	Age	Salary	Location
▶	1	Priya	Bhatia	26	200000	Bengaluru
	2	Suchitra	Behera	24	50000	Bengaluru
	3	Palak	Bhatt	29	90000	Noida
	4	Amit	Agarwal	31	150000	Noida
	5	Avneet	Kaur	23	60000	Bengaluru
	6	Rashmi	Tanwar	29	95000	Gurgaon
	7	Rohan	Kapoor	35	30000	Pune
	8	Priya	Sangwan	26	100000	Pune
	9	Priyanshi	Bhatia	28	10000	Noida
	11	Atul	Bhatt	26	200000	Bengaluru

Bengaluru

per

Window

Noida

Pune

ROW-NUMBER () → assign the

rows to each

Window

RANK ()

it skips the
Rank value

based on how

many rows

DENSE_RANK ()

have same

Ranking

Join, Subqueries, Window functions, CTE